



**Verified Carbon
Standard**

DE AAR HYDRA STORAGE

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1 PROJECT DETAILS

1.1 Summary Description of the Project

De Aar Hydra Storage Project (the "Project") is a single site, dispatchable power facility, comprising of solar photo-voltaic technology with a battery energy storage system (BESS). The Project is located 5km South-East of De Aar in the Northern Cape province of South Africa and roughly 8km North of the Eskom Main Transmission Substation, Hydra. The Project will have a Contracted Capacity of 75 MW (AC) at the Delivery Point. The solar portion is split over three footprints located adjacent to one another and forming a combined total installed capacity of an approximately 216 MW peak (DC). The BESS is to have a combined energy capacity of 495 MWh, and 75 MW dispatch capacity, at the Delivery Point.

Prior to implementing this project electricity was supplied by the current fleet of power plants owned and operated by Eskom, the majority of which are coal-based. The Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is an initiative by the South African government aimed at increasing electricity capacity through private sector investment in solar photovoltaic and concentrated solar, onshore wind power, small hydro, landfill gas, biomass, and biogas [1]. Since coal is an emissions-intensive energy carrier, South Africa's economy is very emissions-intensive, especially when compared to other countries at a similar stage of development as measured by the Human Development Index [2]. South Africa's energy sector is by far the largest contributor to the country's high emissions, contributing 94.7% to the country's predominant GHG, which is CO₂ [3].

Project activity is expected to reduce annual average emission reductions equivalent to 450,000 tCO₂e and 3,150,000 tCO₂e over the crediting period of 7 years.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	Initial Validation is planned for January 2025	VCS	Carbon Check India	To be updated later

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	Scope: 01 - Energy (renewable / non-renewable)
Project activity type	Type I – Renewable Energy

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Project Eligibility

1.4.1 General eligibility

The project is eligible under the scope of the VCS Program since the project activity meets three of the listed points in paragraph 2.1.1 of the VCS Standard, Version 4.7. Table 1 below describes the scope of the VCS program and the project's applicability, and Table 2 further elaborates on this project's eligibility under VCS v4.7 Table 1.

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing coal-based energy supplied by the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies VCS-approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ² which has been approved for use under the VCS Program.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ³ .
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A

² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

Table 2: Excluded Project Activities under the VCS v 4.7 table 1

Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	Not applicable to this project which is not a hydroelectric power plant.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	This is a grid-connected electricity generation activity using solar photovoltaic (PV) and BESS. Hence the project is eligible and not excluded.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Not applicable to this project which is not a waste heat recovery project.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO₂ capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	Not applicable to this project which does combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable to this project which does not combust fossil fuel.

Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO_{2e} per year.	Not applicable to this project which does not replace electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO_{2e} per year.	Not applicable to this project which does not install or replace electricity transmission lines or transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable to this project which does not reduce HFC-23 emissions.

Pipeline listing and validation deadlines - The project activity was listed on the VERRA website in December 2024, before the validation process began. The opening meeting with the VVB will only take place once the project activity has been listed in the project pipeline with a status of 'Under Validation' in the VCS Program project pipeline. Furthermore, validation shall not begin until the 30-day public comment period has begun, and the VVB shall not complete validation until after the 30-day public comment period has ended. As per para 3.8.1 pf VCS standard V4.7, non-AFOLU projects shall complete validation within two years of the project start date and this condition will be fulfilled by this project activity as the start date of the project activity is anticipated to be in 2025 (being when the planned generation will be fed into the national grid and national utility sign-off is achieved) and validation will commence before this date.

The project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved by Verra for use under the VCS Programme for project activities seeking registration with them. ACM0002 is a large-scale methodology and has no capacity limit, and the De Aar Hydra Storage project is a single project in one location and is not part of a larger project or activity.

1.4.2 AFOLU project eligibility

This section is not applicable, seeing that this project is not an AFOLU project.

1.4.3 Transfer project eligibility

This section is not applicable, seeing that this project is not a transfer project or CPA seeking registration.

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

This is not a Grouped project. Hence, this section is not applicable.

1.6 Project Proponent

Organization name	TotalEnergies Hydra Storage (RF) Proprietary Limited
Contact person	Andries Venter
Title	Project Director
Address	21st Floor, Portside, 5 Buitengracht Street, Cape Town, South Africa, 8001
Telephone	+27 82 462 3198
Email	andries.venter@totalenergies.com

Organization name	Anthesis B.V
Contact person	Grant Little
Title	Principal Carbon Project Developer
Address	32 Kloof Street, Gardens, Cape Town, 8001, South Africa
Telephone	+27 21 300 6770
Email	Grant.Little@anthesisgroup.com

1.7 Other Entities Involved in the Project

There are no other entities involved.

1.8 Ownership

TotalEnergies Hydra Storage (RF) Proprietary Limited has the legal right to own and operate the De Aar Hydra Storage facility. There is a Power Purchase Agreement dated 30 November 2023 together with Deed of Lease between the landowners and the project proponent, dated 03 November 2021 which support the ownership of the project activity.

1.9 Project Start Date

Project start date	30-September-2025
Justification	As per the section 3.8 of the VCS standard v4.7, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. Since the project is not yet started the commercial operations, the expected start of commercial operations as per the project schedule included in the signed PPA has been considered as the project start date.

1.10 Project Crediting Period

Crediting period	☒ <i>Seven years, twice renewable</i> ☐ <i>Ten years, fixed</i> ☐ <i>Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)</i>
Start and end date of first or fixed crediting period	30-September-2025 to 29-September-2032

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☐ < 300,000 tCO₂e/year (project)
☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
30-September-2025 to 31-December-2025	150,000
01-January-2026 to 31-December-2026	450,000
01-January-2027 to 31-December-2027	450,000
01-January-2028 to 31-December-2028	450,000
01-January-2029 to 31-December-2029	450,000
01-January-2030 to 31-December-2030	450,000
01-January-2031 to 31-December-2031	450,000
01-January-2032 to 29-September-2032	300,000
Total estimated ERRs during the first or fixed crediting period	3,150,000
Total number of years	7
Average annual ERRs	450,000

1.12 Description of the Project Activity

De Aar Hydra Storage (the "Project") is a single, dispatchable facility, comprising of solar photo-voltaic technology, a battery energy storage system (BESS). The Project is located 5km South East of De Aar in the Northern Cape and roughly 8km North of the Eskom Main Transmission Substation, Hydra. The Project will have a Contracted Capacity of 75 MW (AC) at the Delivery Point. The solar portion is split over 3 x authorized footprints located adjacent to one another and forming a combined total installed capacity of an approximately 216 MW peak (DC). The BESS is to have a combined energy capacity of 495 MWh, 75 MW dispatchable capacity, at the Delivery Point.

An IPP owned 132/33kV substation will be located within the project footprint around 30°40'41.05"S, 24° 3'59.62"E, with an 85MVA 132/33kV Step-up transformer and transformer bay. The isolators on the HV Site of the IPP Transformer is the location of the Main Metering Installation, the Point of Connection (POC) and where energy will be exported from the Project to the Buyer at the Delivery Point. The interconnection works include a 132kV switching station, located adjacent to the IPP Substation, connected to the 132kV network.

Project or Facility Details

Project Name	TotalEnergies Hydra Storage
Contracted Capacity [MW]	75 MW
Design Auxiliary Consumption of the Facility [MW]	Solar,0.72MW. Battery,2.25MW
No. of inverters	Opt. 1: 838 x (String) Preferred Opt. 2: 973 x (String) Alternative Opt. 3: 226 x (Central) Alternative
Type of inverter manufacturer and mode	Option 1: String Inverter – Preferred SunGrow: SG250HX-LN-20 Option 2: String Inverter – Alternative Huawei SUN2000-200KTL-H2 Option 3: Central Inverter – Alternative SunGrow SG1100UD
Maximum instantaneous AC rated capacity of Facility: [MW]	209.500 MW @25degC
Maximum continuous AC rated capacity of Facility: [MW]:	189.735 MW @45degC
Transformer Ratings:	MV:6600 kVA / 8800 kVA 0.66 kV 33/ 0.66kV, Dy11y11, ONAN HV: 85 / 56 MVA 132 kV/33 kV; Three phase, oil immersed, 50Hz YNd1 with an on-load tap changer.
Number of Transformers:	MV: 30 - 37X Hv:1x
Planned voltage connection level:	132kV
Planned connection point (i.e. the name of the substation or distribution line onto which the Facility is intended to connect):	POC: Cluster 1 Switching Station – on the HV Side of the IPP owned Step - up Transformer at the IPP project site. Cluster 1 SS will connect to the Hydra MTS 132kV network
PV Panels	Option 1: Longi (Bifacial 540W-600W – LR5-72HBD-540-600M) Option 2: Seraphim (Bifacial 540W-600W SRP-540-600-BTA-BG) Option 3: Jinko (Bifacial JKM570-590N-72HL4-BDV) Option 4: JASolar (Bifacial: JAM72D40-555-580/GB)
Battery Energy Storage System	Opt.1: Narada (Cell, FE125A, 400Wp Module, 76.8NESP250, 19.2kWh Rack,8 layers, 288kWh)

	Opt.2: SAFT (Cell: LiShen 280Ah LiFePO4 Module: SAFT 21.5 kWh 1P24S) BESS type: SAFT I-Shift 3MWh – 136 modules Opt.3: CATL (CATL: Ener C, 3.72 MWh, 1331.2V) Opt.4: BYD (Cell: C15FHNE 320Ah LiFePO4 Module: S19 300Ah BESS: Cube Pro)
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This is a Greenfield project activity generating electricity from the solar photovoltaic component combined with BESS and supplying it to the national grid. In the baseline scenario the equivalent of electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool 07 “Tool to calculate the emission factor for an electricity system”, version 7.0. There is no technology transfer occurring in the project activity. There was no activity at the site of the project owner prior to the implementation of this project. Hence pre-project scenario and baseline scenario is the same.

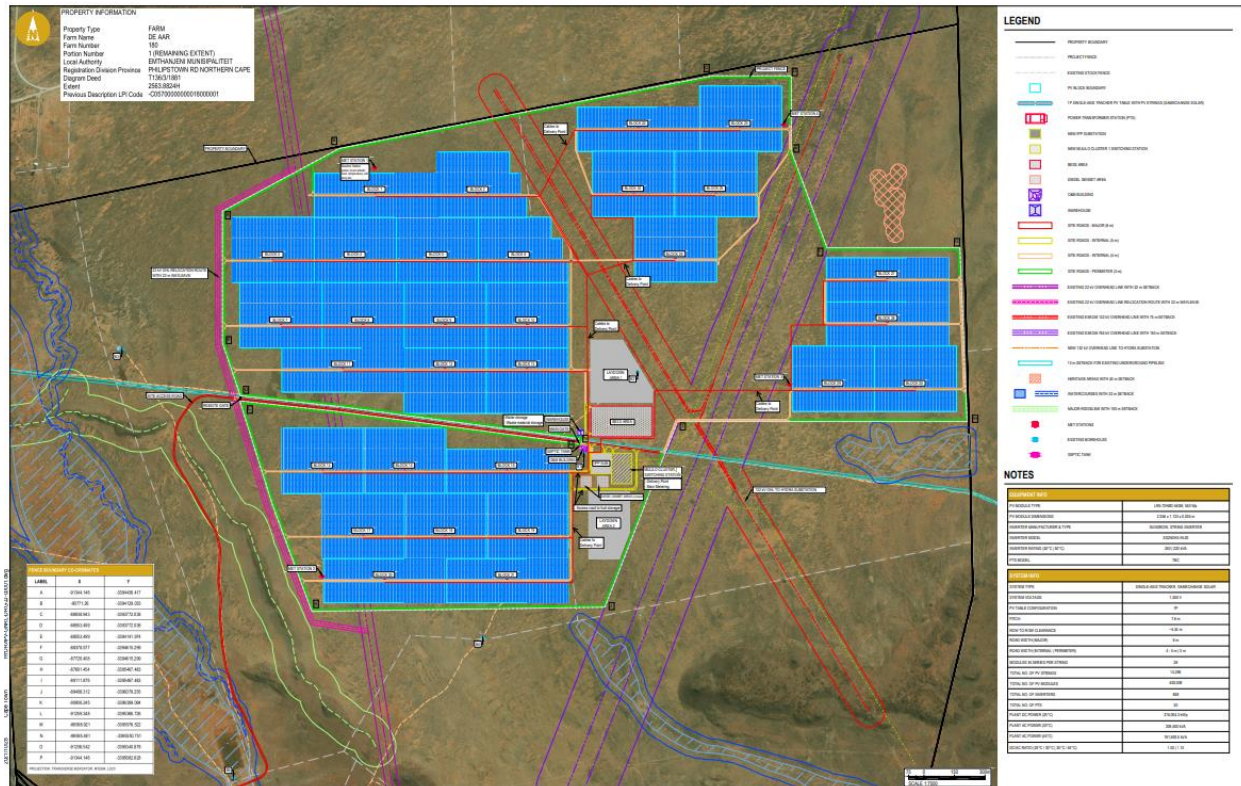


Figure 1: Schematic of proposed project activity site

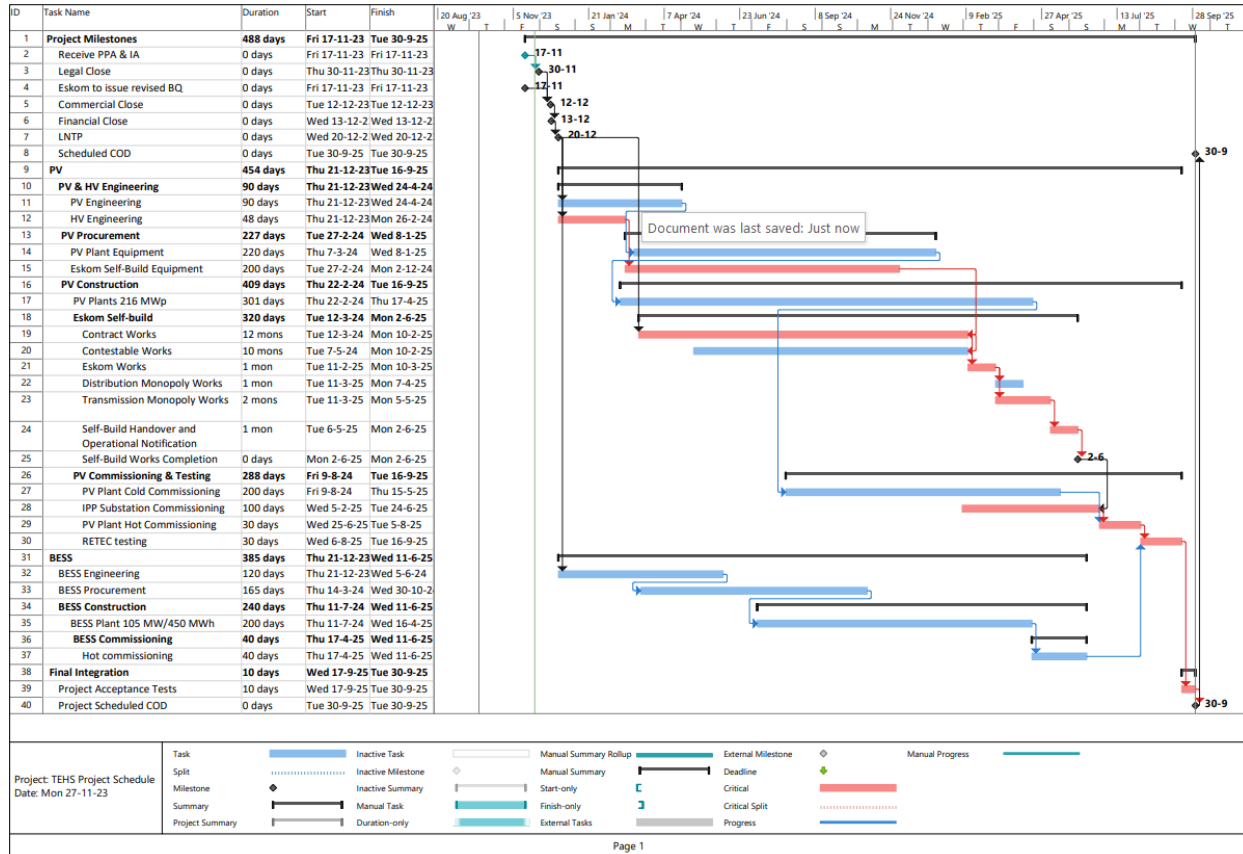


Figure 2: Proposed implementation schedule for the De Aar Hydra project

De Aar Hydra Storage Major Equipment Information:

Equipment	PV Module (35%)	PV Module (65%)	Inverter	BESS	HV Transformer	Tracker
Supplier	Seraphim	JA Solar	Sungrow	CATL	Actom	GameChange
Age	0	0	0	0	0	0
Lifetime	30 Years	30 Years	25 Years	25 Years	25 Years	25 Years
Capacity	75,92 MWp	140,1MWp	184.8 MWac	450 MWh, 105 MW Charging	90 MVA	400 MW

				capacity. 75MW Discharging capacity		
Load Factor	NA	NA	PF at nominal power >0.99	NA	NA	NA
Efficiency	0.4% yearly degradation, PV arrays losses: 0.7 kWh/kWp/ day	0.4% yearly degradation, PV arrays losses: 0.7 kWh/kWp/ day	99%	DC round trip efficiency year 1 >88%, year 2 >87,8%	NA	NA

1.13 Project Location

De Aar Hydra Storage Physical location and geographic coordinates	
Physical address	Portion 1 of farm De Aar No 180 in the Emthanjeni Municipality and vicinity
Latitude	24° 3'59.62"E
Longitude	30° 40'41.05"S

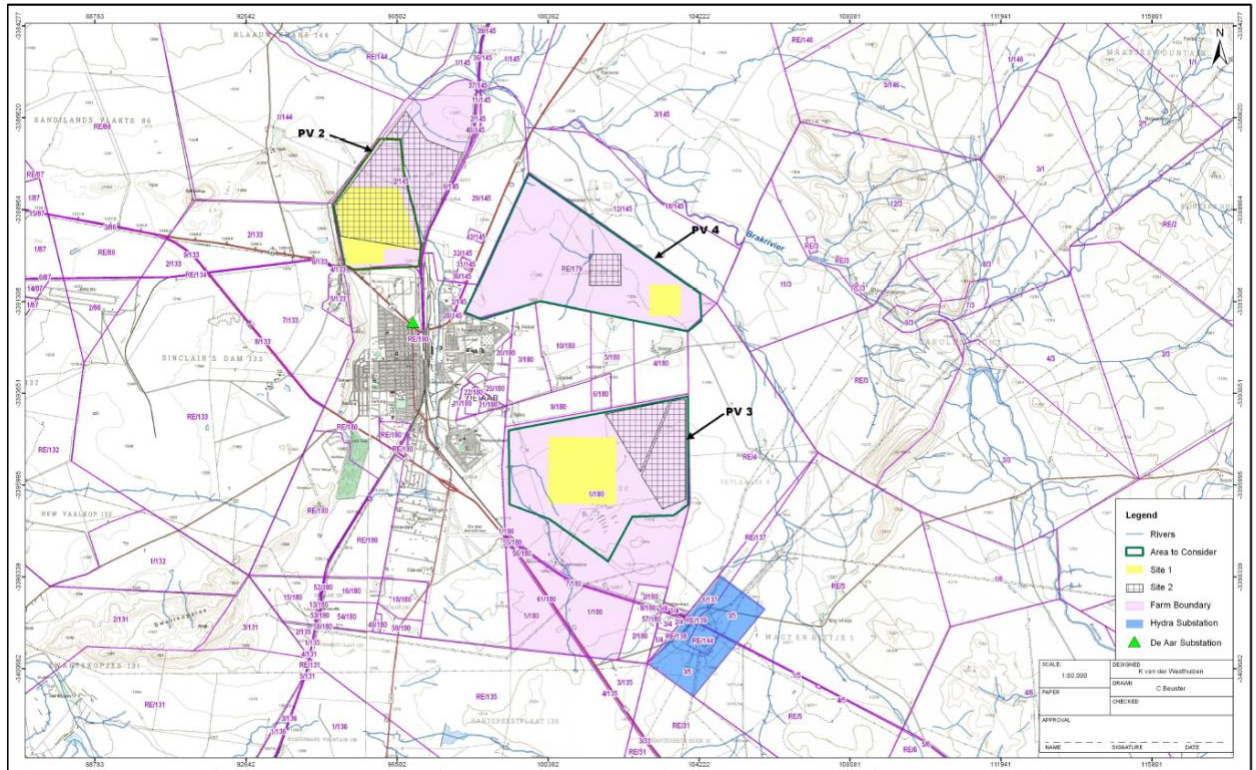


Figure 3: Proposed sites and options for project location

1.14 Conditions Prior to Project Initiation

Prior to implementing this project electricity was supplied by the current fleet of power plants contracted, owned and/or operated by Eskom, the majority of which are coal-based. The Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is an initiative by the South African government aimed at increasing electricity capacity through private sector investment in solar photovoltaic and concentrated solar, onshore wind power, small hydro, landfill gas, biomass, and biogas. Since coal is an emissions-intensive energy carrier, South Africa's economy is very emissions-intensive, especially when compared to other countries at a similar stage of development as measured by the Human Development Index. South Africa's energy sector is by far the largest contributor to the country's high emissions, contributing 94.7% to the country's predominant GHG, which is CO₂.

The primary purpose of the project is to supply renewable and reliable electricity to South African consumers via the national grid, and as such, it is not implemented with the aim of generating GHG emissions for the purpose of their subsequent reduction, removal, or destruction. The project displaces Greenhouse Gas emissions that would be emitted through the burning of coal and other fossil fuels, which is the predominant source of electricity in South Africa. The electricity generation conditions in South Africa, prior to including this plant into the national grid, are further described under Section 3.4 (Baseline Scenario).

Ecosystem type

The project site is located in the Nama Karoo biome which occurs in semi-arid climatic conditions and is dominated by dwarf shrubs and grasses. The ecosystem supports faunal species such as the Black-footed Cat and the Cape Fox which occur in the region for foraging, but not as permanent residents in the area.

Current and historical land-use

The project site is zoned as agricultural land, and much of the land is undeveloped. It is currently used for grazing land since the dominant vegetation type consists of grasses; and the soil characteristics deem it unsuitable for crop cultivation.

Climate

The project site has a semi-arid continental climate that receives approximately 300 mm of rainfall annually over the summer and early autumn seasons. The temperatures range from -8°C to 40°C, and evaporation in the region is approximately 2000 mm annually, which contributes to the arid conditions in the area.

Hydrology

The project is located in proximity to the Brak River, which has its non-perennial tributaries running through the project site. In addition, a wetland was also identified within the project footprint which feeds into the Brak River system. These hydrological features have been classified and approved to be moderately modified, with low ecological significance and sensitivity in the region.

Topography

The topography in the project site is generally flat terrain that slopes gently with an average gradient of less than 10%. The flat terrain makes the area ideal for the proposed development of the solar PV project.

Soils

The soils in the site are described as shallow red apedal soils, which are weakly structured and tend to experience excessive drainage. The soils also have extremely low organic matter content, thus deeming them unsuitable for cultivation.

Vegetation

The vegetation type that occurs in the project site is the Northern Upper Karoo, which is characterized by dwarf karoo shrubs, grasses and low trees. There are also small numbers of alien species present in the project site.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Compliance with provincial, national and international legislation was considered during the planning and implementation of this project. The following legislative aspects were considered:

Table 3: Relevant Legislation

National Environmental Management Act, No. 107 of 1998	PP has the responsibility to ensure that the Project activity conforms to the principles of NEMA. In developing the EIA process, PP has been cognisant of this need, and accordingly the EIA process has been undertaken in terms of NEMA and the EIA Regulations promulgated on 18 June 2010.
National Environmental Management: Biodiversity Act (No. 10 of 2004) (NEM: BA)	The National Environmental Management Biodiversity Act (No.10 of 2004) (NEM: BA) aims to conserve and manage the country's biodiversity via protecting species and ecosystems, specifically those which are threatened and/or are critically endangered.
National Water Act, No 36 of 1998	The National Water Act (NWA) (Act No 36 of 1998) provides for the sustainable and equitable use and protection of water resources. It is founded on the principle that the National Government has overall responsibility for and authority over water resource management, including the equitable allocation and beneficial use of water in the public interest, and that a person can only be entitled to use water if the use is permissible under the NWA. Section 21 of the NWA specifies the water uses which require authorisation from the Department of Water Affairs (DWA) in terms of the NWA before they may commence.
National Heritage Resources Act, No. 25 of 1999	In terms of the National Heritage Resources Act (No. 25 of 1999) (NHRA), any person who intends to undertake "any development ...

	which will change the character of a site exceeding 5 000 m2 in extent”, “the construction of a road...powerline, pipeline...exceeding 300 m in length” or “the rezoning of site larger than 10 000 m2 in extent...” must at the very earliest stages of initiating the development notify the responsible heritage resources authority, namely the South African Heritage Resources Agency (SAHRA) or the relevant provincial heritage agency. These agencies would in turn indicate whether a full Heritage Impact Assessment (HIA) would need to be undertaken.
Conservation of Agricultural Resources Act, No. 43 of 1983	The Conservation of Agricultural Resources Act (No. 43 of 1983) (CARA) makes provision for the conservation of the natural agricultural resources of South Africa through maintaining the production potential of land, combating and preventing erosion, preventing the weakening or destruction of the water sources, protecting vegetation, and combating weeds and invader plants. Regulation 15 of CARA lists problem plants (undesired aliens, declared weeds, and plant invaders). Plants listed in this regulation must be controlled by the landowner.
Policies regarding greenhouse gas and carbon emissions;	Electricity generation using carbon-based fuels is responsible for a large proportion of carbon dioxide (CO₂) emissions worldwide. In Africa, the CO₂ emissions are primarily the result of fossil fuel burning and industrial processes, such as coal fired power stations. South Africa accounts for some 38 % of Africa’s CO₂ emissions. The global per capita CO₂ average emission level is 1.23 metric tonnes. In South Africa however, the average emission rate is 2.68 metric tonnes per person per annum. The International Energy Agency (2008) “Renewables in global energy

	supply: An IEA facts sheet” estimates that nearly 50% of global electricity supplies will need to come from renewable energy sources in order to halve carbon dioxide emissions by 2050 and minimise significant, irreversible climate change impacts.
White Paper on the Energy Policy of the Republic of South Africa (1998);	As required by the Constitution of the Republic of South Africa (Act No. 108 of 1996), the White Paper on the Energy Policy of the Republic of South Africa (1998) was published by the Department of Minerals and Energy in response to the changing political climate and socioeconomic outlook. Key objectives are identified in terms of energy supply and demand, as well as co-ordinated with other social sectors and between energy sub-sectors. While considering the larger environmental implications of energy production and supply, the White Paper looks into the future to adopting an integrated resource planning approach, integrating the environmental costs into economic analysis. It is with this outlook that the renewable energy, including solar energy, is seen as a viable, attractive and sustainable option to be promoted as part of South Africa’s energy policy towards energy diversification.
White Paper on Renewable Energy (2003);	Published by the Department of Minerals and Energy (DME) in 2003, the White Paper on renewable Energy supplements the above-mentioned Energy Policy which identified the medium- and long-term potential for renewable energy as significant. The White Paper sets out the vision, policy principles, strategic goals, and objectives in terms of renewable energy. The White Paper presents South Africa’s options in terms of renewable energy as

	extensive and a viable and sustainable alternative to fossil fuel options. A strategic programme of action to develop South Africa's renewable energy resources is proposed, particularly for power generation and reducing the need for coal-based power generation.
National Energy Act (No. 34 of 2008) and Electricity Regulation Act (ERA) (No. 4 of 2006);	South Africa has two acts that direct the planning and development of the country's electricity sector. In terms of the New Generation Regulations, the Integrated Resource Plan (IRP) has been developed by the DoE and sets out the new generation capacity requirement per technology, taking energy efficiency and the demand-side management projects into account. This required, new generation capacity must be met through the technologies and projects listed in the IRP and all IPP procurement programmes will be undertaken in accordance with the specified capacities and technologies listed in the IRP
IPP (Independent Power Producer) Procurement Process;	The introduction of private sector generation thus has multiple benefits. It will contribute greatly to the diversification of both the supply and nature of energy production, assist in the introduction of new skills and capital into the industry, and enable the benchmarking of performance and pricing.
Integrated Energy Plan for the Republic of South Africa (2003); and	Commissioned by DME in 2003, the Integrated Energy Plan (IEP) aims to provide a framework in which specific energy policies, development decisions and energy supply trade-offs can be made on a project-by-project basis. The framework is intended to create a balance in providing low-cost electricity for social and economic developments, ensuring security of supply,

	and minimizing the associated environmental impacts.
Integrated Resources Plan.	The Integrated Resource Plan (IRP) is a National Electricity Plan, which is a subset of the Integrated Energy Plan. The IRP is also not a short or medium-term operational plan but a plan that directs the expansion of the electricity supply over the given period.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. In alignment over the seven-year crediting period, renewable twice, the project anticipates an average annual emission reduction of 450,000 tons of CO₂ equivalent, totalling 3,150,000 tons of CO₂ equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This aligns with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 452 000 MWh per annum of emission-intensive grid electricity through the generation of low-emission renewable electricity.

Furthermore, aligned with SDG 8.5, the project will promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. This will be achieved through just less than 900 job opportunities during the construction phase and 90 jobs during the ongoing operation of the project into the future.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7,2,1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 452 000 MWh per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 9 400 GWh of emission intensive grid electricity through the generation of low emission renewable electricity.
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 450 000 tCO ₂ e per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 9,450,000 tCO ₂ e of emission intensive grid electricity through the generation of low emission renewable electricity over the 21-year project lifetime.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 24.9% to 33.5% over the past decade (2012 to 2022) this SDG target remains a high priority for the country. The country has a target to, “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	During the construction phase there will be 890 job opportunities created and during Operation there will be approximately 90 jobs. Furthermore, the project has committed to contributing more than 40% of the capital expenditure to local content (procurement of South African goods and services).	The project planned to create 890 job opportunities during the construction phase (from domestic labour) and a further 90 permanent jobs during operations. The Operational jobs will be ongoing, but the construction jobs would only be during the construction phase and will phase out over a period once the plant is operational. Over the 20-year operational period, 1% of revenue will be dedicated to local communities through socio-economic initiative.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per the applied methodology ACM0002 (Ver. 22.0), no leakage emissions are considered in the project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The approach adopted in the stakeholder identification process was to contact as many stakeholders as possible through several channels, which included:

- Placing advertisements in a local De Aar newspaper, The Echo, on 4 November 2011 and in the provincial newspaper, Die Volksblad, on 3 November 2011. A further advertisement was run in December 2024 in a local newspaper.
- Placing a notice board at the site and at strategic locations in the vicinity of the project with a request for feedback.
- Providing written notice to potential stakeholders, including surrounding landowners, organs of state and relevant authorities; and
- Requesting potential stakeholders to recommend other potential stakeholders to include them on the database (chain referral process).

The stakeholder Register is continuously updated as new I&APs are identified and/ or requested to be included or excluded in the database as part of the engagement process. The stakeholder register is maintained centrally.

All stakeholder details were compiled into a database which includes contact details, dates and details of consultations, and a record of all issues raised, and responses provided. This database will be updated on an on-going basis.

Authorities regarded as key stakeholders:

- Northern Cape DEANC.

	· Department of Agriculture, Forestry and Fisheries: Directorate Land Use and Soil Management. · Northern Cape: Department of Agriculture · Department of Water Affairs (DWA): Deputy Director Lower Orange WMA. · South African Heritage Resources Agency. · Northern Cape Provincial Heritage: Boswa ya Kapa Bokone; · Pixley ka Seme District Municipality. · Emthanjeni Local Municipality; and · Eskom.
Legal or customary tenure/access rights	The legal tenure of the land is private ownership, and a binding lease agreement is in place with the landowners, in order to establish the De Aar Hydra Storage plant facilities on the land.
Stakeholder diversity and changes over time	Key stakeholders include government agencies, neighbouring landowners and residents, local communities, businesses and environmental NGOs. The project site lies within the Emthanjeni Local Municipality in the Northern Cape, wherein social issues include unemployment and poverty. Wool production and livestock farming contribute largely to the economy in De Aar.
Expected changes in well-being	No significant negative impacts on the well-being of stakeholders are anticipated to occur because of this project. However, there are many positive impacts. Over the 20-year operational period, the local community will benefit from a community trust with an equity share in the project life to benefit historically disadvantaged communities. The project proponents will also initiate a training strategy to enable employment from the local community, and also prioritize local suppliers of components required in the construction phase.

Location of stakeholders	Nonzwakazi is the closest local community that may be influenced by the project activity. Residents of De Aar (within the Emthanjeni Local Municipality) may also be affected by the project implementation.
Location of resources	The project is located primarily on the following location: Remaining Extent of Portion 1 of the Farm De Aar No 180 in the Emthanjeni Municipality, Division Philipstown, Province Northern Cape

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	30 November 2011 (initial)
Stakeholder engagement process	Stakeholders that were identified and those who provided feedback from the newspaper article and call for engagement were notified about the focus group meeting which was held at the local community hall in De Aar. The meeting was open for all interested parties to attend and was held to provide feedback to the stakeholders about the stakeholder engagement process and comments that were received by the project proponent. The meeting also provided the stakeholders with an opportunity to comment on the findings presented and to raise any questions or uncertainties. Further stakeholder consultation was help by sending emails to the entire stakeholder database prior to validation and also having notices in key public areas in the vicinity of the project. A newspaper advertisement was also placed and stakeholders were encouraged to provide consent and comments to a dedicated email provided by the project proponent. The communication was presented in a culturally appropriate and inclusive manner, and inclusive language was used.
Consultation outcome	The final comments and feedback received from this engagement were included in the final EIA report and were taken into

	consideration in the development of the Environmental Management Programme for the project.
Ongoing communication	Ongoing communication via direct electronic communication to stakeholders recorded in the registered I&AP database, as well as public notices placed in accessible locations in the surrounding communities. In addition, community liaison officers have been appointed to facilitate communication with local communities on an ongoing basis. All grievances received are recorded and addressed as far as practically possible. Update meetings have also been held in the surrounding communities to keep the stakeholders informed about the progress of the project. The last round of community meetings was held in July 2024. The project proponent will keep a register of communication from stakeholders in relation to the project and any resultant actions or review thereof.
Stakeholder input	All comments received during the EIA and stakeholder consultation has been carefully considered and addressed in the EIA and project implementation. No updates to the project design have been considered necessary. Stakeholder comments will continue to be monitored to consider whether further updates to the project are required.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The land lease between landowners and the Operating project proponent was signed in September 2021. No IPs (Indigenous Peoples), LCs (Local Communities) or customary rights holders are impacted by the De Aar Hydra Storage project activities.
Outcome of FPIC	The outcome of the process is that the De Aar Hydra Storage facility will be established on private land for the next 25 years with option to renew up to another 25 years. No other land resources or communities have been negatively impacted.

2.1.4 Grievance Redress Procedure

Development process	Grievances can be received via two channels: - The grievance portal on the Anthesis website - There are also permanently employed community liaison officers stationed in the surrounding communities with the goal of keeping communication up-to-date with the De Aar Hydra Storage project. Any grievances received by Anthesis will be handled on a case-by-case basis by Anthesis in consultation with the De Aar Hydra Storage project team. Those received by the De Aar Hydra Storage team will be communicated with Anthesis periodically.
Grievance redress procedure	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. The Grievance Redress process consists of: - Receiving the issue from a stakeholder - Assigning it to the relevant manager/team - Responding to the stakeholder (which may include telephonic, written or face to face communication) in order to understand and resolve the issue in good faith - Recording the outcome of the interaction. - Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third party and failing resolution shall be raised to either arbitration or courts of law in South Africa. Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged, reviewed and taken into account in ongoing project operation, management reporting and future verifications.

2.1.5 Public Comments

Comments received	Actions taken
This section will be completed once the public commenting period has passed. No comments received at the time of publishing.	This section will be completed once the public commenting period has passed.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards aligned with the VCS requirements. Members of the project development team have as a collective, in excess of 50 years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits a company for over 2 decades.

The De Aar Hydra Storage operational project proponent has extensive renewable energy project experience in the South African region and globally, including similar Renewable Power facilities in Prieska, South Africa, Abu Dhabi, UAE and France among others. The integrated teams are well-positioned to implement and operate the facility and deliver the associated GHG emission reductions into the local market.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No natural and human induced risks	Measures to prevent natural and human-induced risks were taken and documented in the EIA report.
Risks to stakeholder participation	No risks identified	The stakeholder engagement process was conducted, and comments received from the stakeholders were properly addressed. No evidence of participation barriers to stakeholders identified.

		The process for stakeholder's participation was undertaken as required by the South African EIA requirements.
Working conditions	No risks identified related to working conditions	All workers and their working conditions are subject to the Basic Conditions of Employment Act of South Africa and labour conditions are covered by periodic inspections and compliance to the South African Department of Labour.
Safety of women and girls	No risks related to women and girls were identified	The project proponent has taken all measures to ensure the safety of women and girls by implementing various policies as per South African labour and anti-discrimination laws.
Safety of minority and marginalized groups, including children	No risks were identified	The project proponent ensures the safety of minority and marginalized groups, including children in the project as per the South African labour and anti-discrimination laws.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Increased levels of air and noise pollution are anticipated during the construction phase due to higher traffic volumes.	- Speed of vehicles on roads should be controlled (e.g. speed bumps and speed restrictions.) - All roads should preferably be sealed to eliminate dust formation caused by strong winds and vehicle movement.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁴	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	The project proponents have compiled human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Sexual harassment	No risks identified	The project proponents have compiled Formal human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Equal pay for equal work	No risks identified	The project proponents have compiled human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Gender equity in labor and work	No risks identified	The project proponents have compiled human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Forced labor	No risks identified	The project proponents have compiled human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
Child labor	No risks identified	The project proponents have compiled human resource and labor policies as per Basic Conditions of Employment Act of South Africa.

⁴ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Human trafficking	No risks identified	The project proponents have compiled human resource and labor policies as per Basic Conditions of Employment Act of South Africa.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project proponent adheres to international and national human right laws and the Bill of Rights entrenched within the Constitution of South Africa. Risks related to human rights were assessed during the EIA process.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Western boundary of project site is adjacent to a graveyard; and development beyond the boundary may have an effect on the graveyard.	Buffer zone must be maintained along western boundary of the project site to ensure minimisation of disturbance and risk for the graveyard.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	All property rights are covered by legally binding lease agreements that cover land tenure, and no at-risk communities were involved or affected.

2.3.5 Benefit Sharing

The project does not impact on property rights and therefore no benefit sharing agreement is required.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	Disturbance and displacement of mammal species from foraging and breeding areas.	Restrict construction footprint to a bare minimum and place demarcations for no-go areas to minimize disturbance impacts
	Mortality of avifaunal species from collisions with new powerlines.	Develop a bird monitoring programme for population surveys and collision assessments
Soil degradation and soil erosion	Clearance of land during construction increases the risk of soil erosion.	Site must be cleared in sections, and rehabilitation must incorporate coverage of bare soil with straw and local indigenous vegetation.
Water consumption and stress	Increased risk of sedimentation of surrounding water courses	Construction should take place during the low flow period to minimize the risk of erosion and contaminated runoff from construction site into freshwater systems.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

Species and habitat	There is no significant risk to species or habitats that had been identified.
Areas needed for habitat connectivity	The project has not adversely impacted any areas of significance that are needed for habitat connectivity

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	No mitigation /preventive measure taken as there is no threat to Areas for habitat connectivity
Areas for habitat connectivity	No risk identified	No mitigation /preventive measure taken as there is no threat to Areas for habitat connectivity

2.4.2 Introduction of Species

Not applicable, as there is no introduction of new species in the construction or operation of the project activity.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
<i>Opuntia ficus-indica</i>	Immediate control measures must be implemented that are in line with established norms and legal limitations. Project site must be fenced off to limit movement in the site and develop an ongoing invasive species monitoring programme.
<i>Agave americana</i>	Immediate control measures must be implemented that are in line with established norms and legal limitations. Project site must be fenced off to limit movement in the site and develop an ongoing invasive species monitoring programme.
<i>Prosopis glandulosa</i>	Immediate control measures must be implemented that are in line with established norms and legal limitations. Project site must be fenced off to limit movement in the site and develop an ongoing invasive species monitoring programme.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risks identified	No invasive species are to be introduced. Measures taken are documented in the EMPr.

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	This project is not an ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system	7.0
Tool	TOOL 24	Common practice	3.1
Tool	TOOL 27	Tool for the Investment Analysis	14.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s) (c) Involve a retrofit of (an) existing operating plant(s)/unit(s);	The project activity involves installation of a greenfield power plant that is connected to the national grid of South Africa, so, this meets the applicability condition (a).

	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP.	This project activity is a grid-connected renewable energy power generation project which includes the integration of a BESS into the greenfield power plant. So, the project meets applicability condition (a)

	The BESS is located at site of the greenfield renewable power plant.	
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may	(a) This project is a solar renewable energy plant utilising solar power. (b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant. (c) BESS is an integral part of the project design and is a requirement of the project approvals and power purchase

	the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	agreement⁵ with the national grid. (d) BESS will only be charged from the greenfield renewable energy plants at each location. No grid power will be used to charge the BESS component. (e) Not applicable to this project as the project does not employ PSP
	In case of hydro power plants, one of the following conditions shall apply:	Not applicable to this project as the project

	(a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total	does not employ hydro power
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	installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from rivers and tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Not applicable to this project as the project does not employ hydro power
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project as the project does not employ PSP
	The methodology is not applicable to:	(a) Not applicable to this project

	(a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site. (b) Biomass fired power plants/units.	as the project does not employ any fuel switching (b) Not applicable to this project as the project does not employ any biomass firing
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable to this project as the project does not employ any retrofits, rehabilitations, replacements, or capacity additions

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Project activity does not entail proposing a new methodology. As such TOOL 01 is mandatory, and therefore was applied to prove additionality.

CDM TOOL 01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	TOOL 01 is referenced in the methodology ACM0002 as the prescribed tool to prove additionality and, as such it is mandatory for this project.
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Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 07	This tool may be applied to estimate the Operating Margin (OM), Build Margin (BM) and/or Combined Margin (CM) when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This project will feed electricity into the South African national grid, and as such, this is the prescribed tool to calculate the baseline emissions.
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants or, as an option, can include off-grid power plants. In the latter case, two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power	This project does not include any off-grid power plants. Therefore, this tool was used to calculate the emission factor for the project baseline electricity system for grid power plants only.

	plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
CDM TOOL 07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in a non-Annex I country and hence the tool is applicable.
CDM TOOL 07	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 24	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Project activity applies methodological tool: Tool for the demonstration and assessment of additionality and therefore the Common Practice tool is applicable to the project activity.
CDM TOOL 24	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Methodology ACM0002 does not define separate approaches for the conduction of common practice test, which are different from the ones on the tool.

Methodology ID	Applicability condition	Justification of compliance
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CDM TOOL 27	This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	Project activity complies as it does use the investment analysis to demonstrate additionality. This is presented in §3.5 below.
CDM TOOL 27	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Project activity complies as methodology ACM0002 provides for the use of investment analysis.

3.3 Project Boundary

As per the methodology ACM002 version 22, the project boundary includes the power plants, and all power plants connected physically to the electricity system that the project power plant is connected to.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid-connected electricity generation	CO ₂	Yes	Main emission source (Grid electricity)
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	
		Other	No	No other GHG emissions relevant.
Project	Greenfield solar photovoltaic electricity generation project activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Charging of BESS using electricity	CO ₂	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

Source	Gas	Included?	Justification/Explanation
from the renewable energy plants	CH ₄	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
	N ₂ O	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
	Other	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

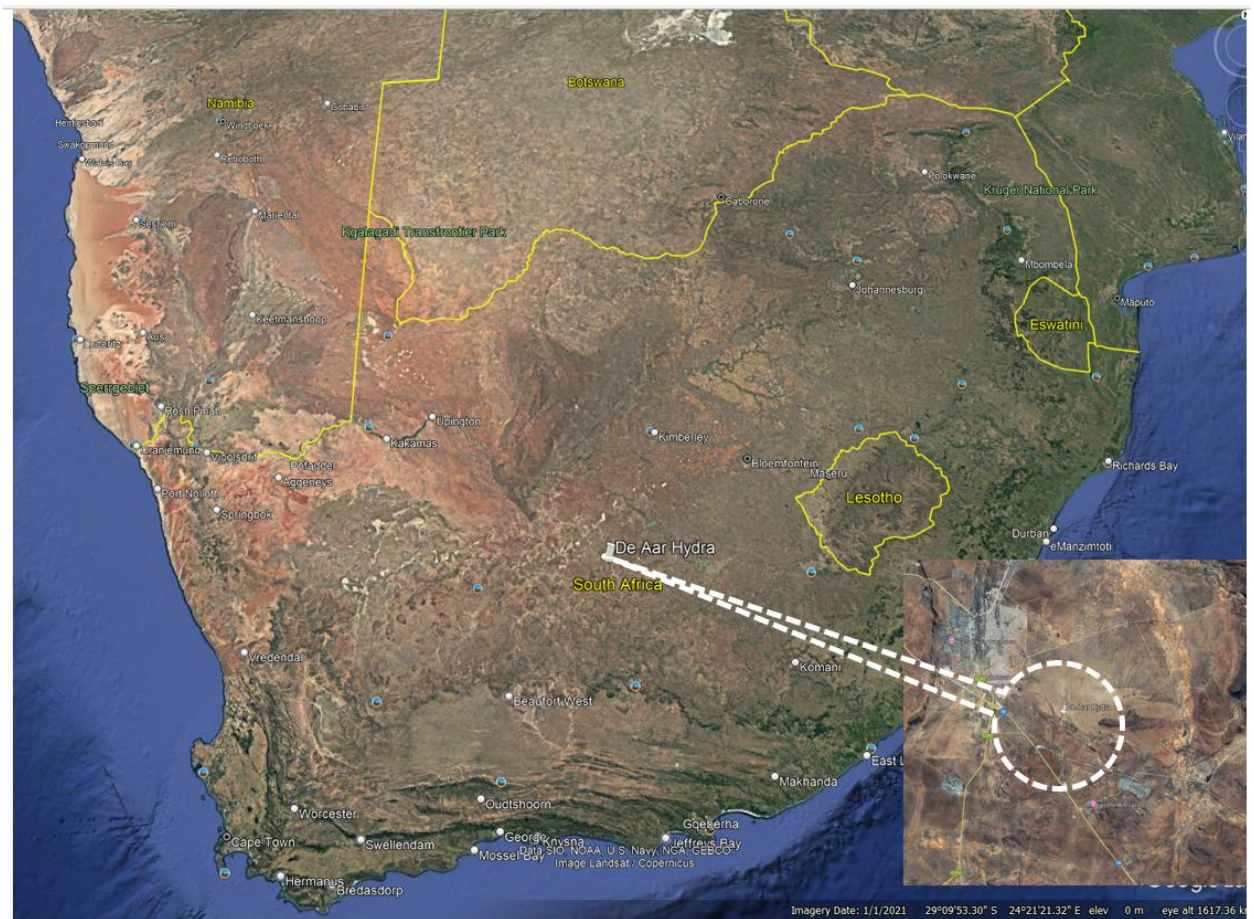


Figure 4: Physical location of the project activities

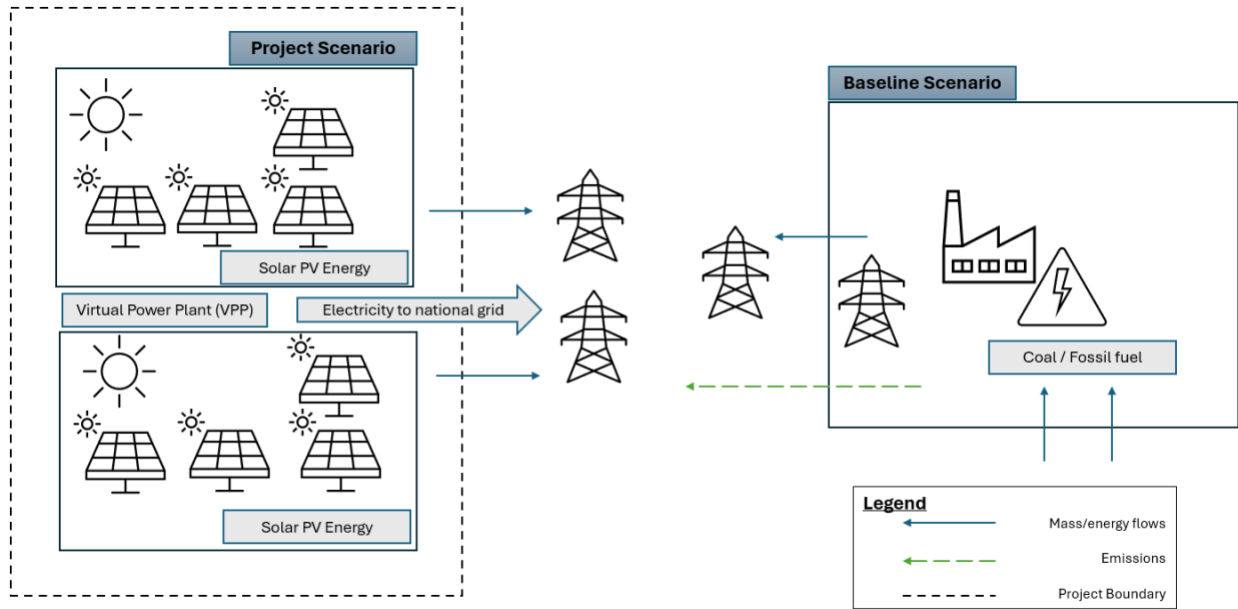


Figure 5: Project boundary, mass & energy flows of De Aar Hydra Project

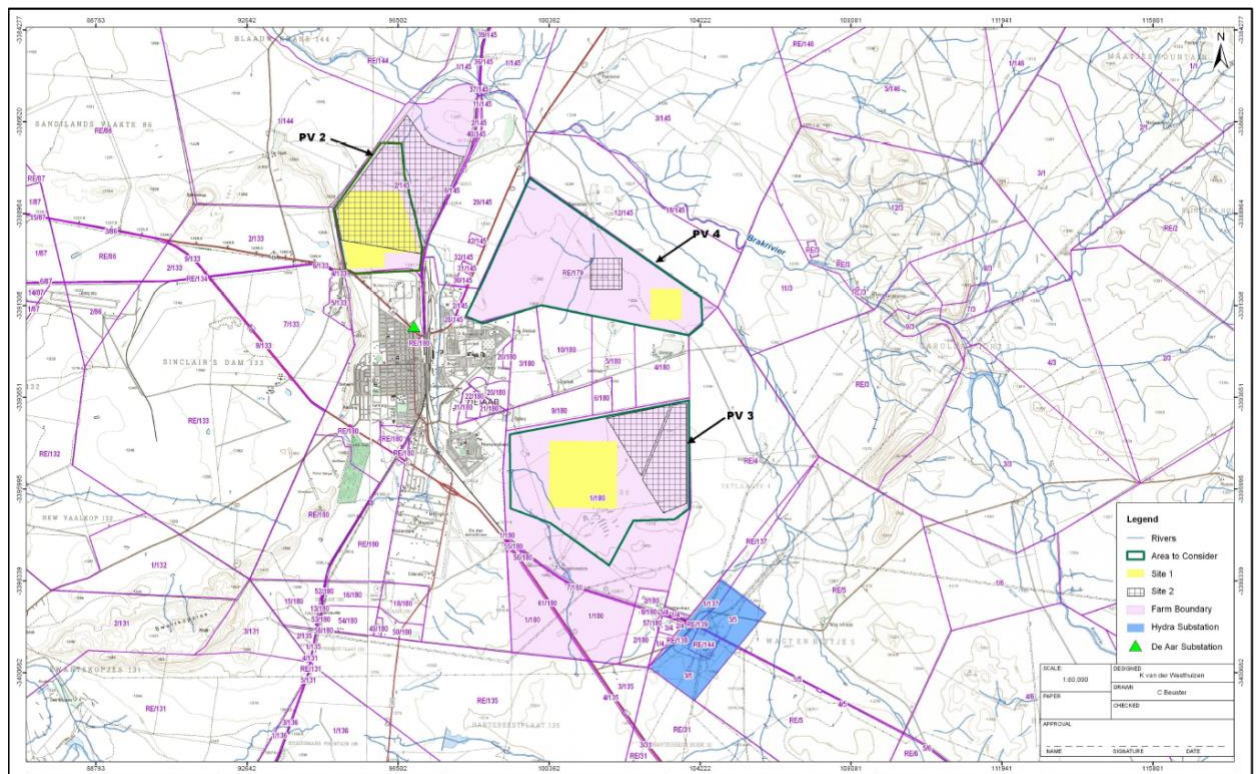


Figure 6: Proposed locations of project activities

3.4 Baseline Scenario

The baseline scenario for the De Aar Hydra Storage project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel power plants, predominantly coal, and the addition of new fossil fuel-based generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: “If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL07 ‘Tool to calculate the emission factor for an electricity system (Version 07.0)’.

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix [1]. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally [2].

According to South Africa’s Integrated Resource Plan (IRP 2019) [3], coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite government efforts to increase renewable energy through initiatives like the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), the expansion pace is slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation. The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.959 tCO₂e/MWh over the past five years, driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the de Aar Hydra project, further supporting the selection of this baseline scenario. The grid emission factor for this baseline is calculated in accordance with the CDM TOOL07 “Tool to calculate the emission factor for an electricity system” Version 07.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.959 tCO₂e/MWh. The steps taken to calculate the grid emission factor as per AM Tool 07 are

outlined in section 4.1 below. §28 – 34 of ACM0002 do not apply to this project, as shown in Table 4 below:

Table 4: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant. The BESS components are also greenfield and exclusive to the greenfield De Aar Hydra generation and storage facilities.
5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind power plant/unit with the integration of BESS	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.
5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield facility and is not connected to a Greenfield pumped storage power plant.

The project activity is the installation of a greenfield solar PV power plant with associated BESS component. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.

3.5 Additionality

Additionality was demonstrated as stipulated in section 5.3 of ACM0002, using CDM tool01 version 7. In addition, Tool 27 version 14 and tool 24 version 3.1 were applied.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

N/A – the project is located in the Republic of South Africa which is a Non-Annex 1 country and activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Additionality is demonstrated by proving that the emission reductions achieved by the project would not have occurred in the absence of carbon finance. Tool 27 version 13 prescribes a stepwise approach, as follows:

The input values used in the investment analysis were valid and applicable at the time of the investment decision taken by the project participant. For this project, financial close was reached in December 2023, and the investment analysis was therefore carried out with 2023 values. The steps followed are outlined below:

Step 0: Demonstrate whether the proposed activity is the first-of-its-kind.

This is not a first-of-its-kind project as there are other solar PV and BESS projects of a similar scale within the country.

Step 1: Identification of alternatives to the project activity

Step 1 requires that realistic and credible alternatives to the project activity be defined through the following sub-steps:

Sub-Step 1a: Define alternatives to the project activity

This step entails defining realistic and credible alternatives available to the project participants or similar project developers that provide outputs or services comparable to the proposed project activities. This is shown in the table below:

Project Alternatives	Applicability to this project
Alternative a: The proposed project activity is undertaken without being registered as a VCS project	Realistic scenario: The project proponent could proceed with this project without carbon finance. The project would still produce renewable energy for sale into the South African electricity grid. This is legally and practically an option, but financial risk associated with the project will be increased thereby rendering it financially not a viable alternative.
Alternative b: Other realistic and credible alternative scenarios to the proposed project activity scenario that delivers electricity with comparable quality, properties and application areas, taking into account examples of scenarios identified in the underlying methodology (ACM0002).	Not a realistic scenario: The generation of renewable energy at the project sites from other sources such as hydro, biomass, geothermal or tidal. The project location is specifically suited to solar and therefore the other alternatives were not considered viable.
Alternative c: Continuation of the current situation (no project activity is undertaken.) This scenario is therefore equal to the baseline scenario where electricity is delivered to the grid through carbon-intensive fossil-fuel derived energy.	Realistic scenario: The grid could continue operating from fossil-fuel powered plants as described in the baseline scenario.

Outcome of sub-step 1a:

Alternative a: The project proponent could not proceed with this project without carbon finance.

Alternative c: Continuation of the current situation (i.e. baseline scenario).

Sub-Step 1a: Consistency with mandatory laws and regulations

The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to

mitigate local air pollution. (This sub-step does not consider national and local policies that do not have legally binding status.)

The scenarios identified in sub-step 1a include solar, wind, and fossil-fuel derived electricity currently feeding into the South African grid. All three these technology options are in compliance with all legal and regulatory requirements in South Africa. There are no laws prohibiting the implementation of the scenarios identified, and other projects do exist in the country as mentioned in Step 0 above. In addition, all applicable environmental laws have been adhered to as indicated by the approved environmental and water use authorisations.

Outcome of sub-step 1b:

The assessment indicates that scenarios listed above are in compliance with mandatory legislation and regulations in South Africa.

The project proponent can now proceed to step 2 or step 3. Project proponents may also select to complete both steps 2 and 3.

Step 2: Investment analysis

This step entails determining whether the proposed project is not:

- (a) The most economically or financially attractive; or
- (b) Economically or financially feasible, without the revenue from the sale of certified emission reductions (CERs).

The latest version of the “Guidelines on the assessment of investment analysis”, was also considered during this step. Option (b) was selected for analysis which is to conduct a financially attractive option. The following sub-steps were applied:

Sub-step 2a: Determine appropriate analysis method

Determine whether to apply simple cost analysis, investment comparison analysis or benchmark analysis. The scenarios identified in step 1 generate financial benefits and therefore the simple cost analysis is not suited. The alternative of continuing the current scenario does not involve an investment by the project developer and therefore the investment comparison analysis is not suited. The benchmark analysis (option III) was most suitable and therefore used for this analysis.

Sub-step 2b: Apply Investment benchmark analysis

Identify financial indicator:

According to Tool 01, version 7, either project IRR or equity IRR can be used. The financial indicator identified as most appropriate for this type of project was identified as pre-tax project IRR. This is a parameter that is used commonly within the market. According to Tool 27 §16, the applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rates or Weighted Average Cost of Capital (WACC) are appropriate benchmarks for a project IRR. Local commercial lending rate

was chosen as a suitable benchmark, as obtained from the South African Reserve Bank at the time of investment decision. For 2023, a value of 11.75% was used[4].

Sub-step 2c: Calculation and comparison of financial indicators

The project IRR benchmark analysis was carried out based on the financial assumptions made during the project decision making process. Based on the results, the project IRR is assessed to be less than the selected benchmark. This substantiates that the investment is not financially attractive (Project IRR for the project activity is less than the Benchmark). Therefore, it can be concluded that the project activity is additional & is not the business-as-usual scenario.

The underlying assumptions are tabulated below:

Project Parameter	Value	Unit	Source
Total Capacity	75	MW	PPA [5]
Annual generation	452,000	MWh/yr	Production model (average over 20 years)
Deration in 1st year	0.3%	%	Production model
Deration 2nd year onward	0.054%	%	Production model
Total plant Capex	7,050,800,000	ZAR	Project Financials
Tariff rate	1678	ZAR/MWh	PPA (weighted average)
O&M Costs	59,863,000	ZAR/yr	Average over 20 years. Supplier contracts, inflated annually by CPI
Escalation On O&M and insurance	5.1	%	Based on supplier agreements and on average CPI at time of investment decision which was in 2023 [6],[7]
Escalation on insurance	5.25%		Based on supplied agreement
Escalation On tariff	4.5	%	Based on South Africa's CPI ⁶ at base date for Power Purchase Agreement, which was 2020 [8],[5]
Insurance	30,631,000	ZAR/yr	Supplier contracts, 1 st year, inflated annually by CPI
Other recurring costs	46, 700, 000	ZAR/yr	Inflated annually by CPI
Salvage value	-80, 037, 000	ZAR	Decommissioning report
Commercial lending rate	11.75	%	SA Reserve Bank[4]

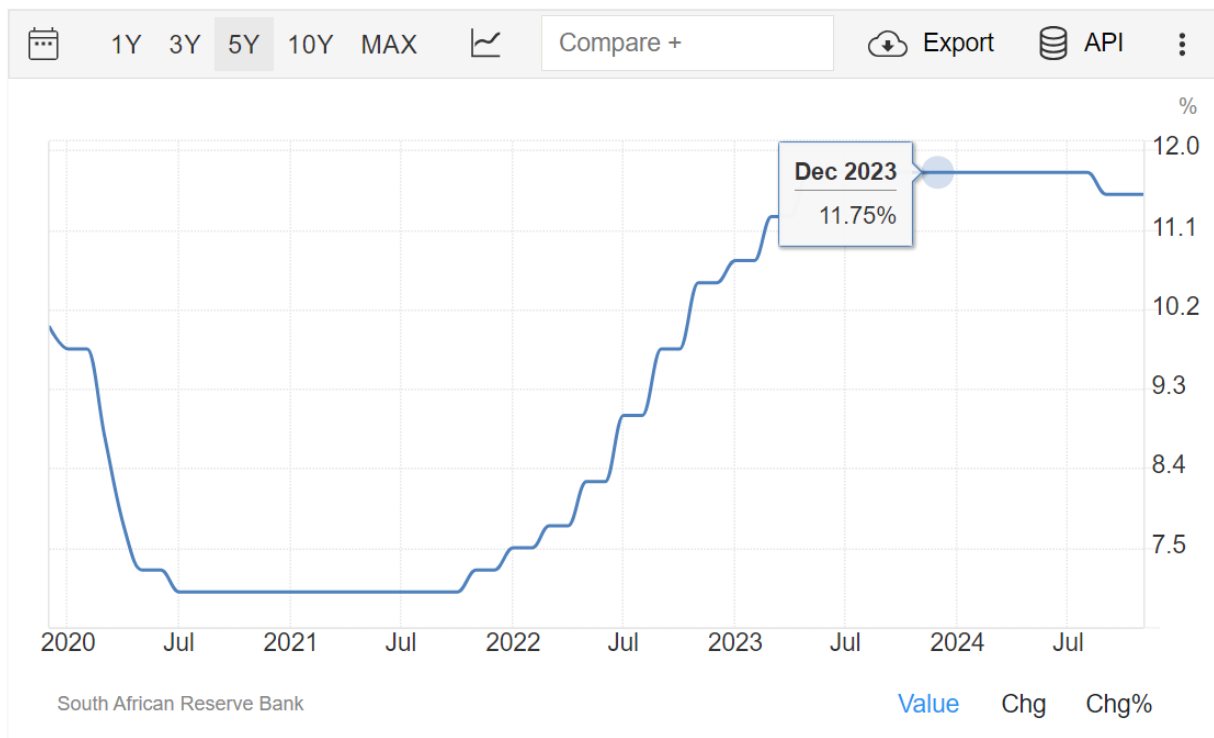
⁶ CPI = Consumer Price Index

Based on the assumptions above, the pre-tax project IRR would be 9.86 % which is below the benchmark of 11.75%

The graph below shows the CPI for South Africa at the time of investment decision[7]:



The commercial lending rate for South Africa in December 2023 is shown in the diagram below [4]:



Sub-step 2d: Sensitivity Analysis

A sensitivity analysis was conducted to demonstrate that the project IRR value remains below the benchmark threshold, even when there are variations to the critical input assumptions, as required by the Tool. A range of +10% and -10% was applied. The following variables are considered in the sensitivity analysis:

- Capital cost
- Operations and maintenance costs
- Electricity tariff
- Annual generation

Parameter	Project IRR at -10%	Project IRR at +- 0%	Project IRR at + 10%
Capex	11.2%	9.86 %	8.71%
O&M	8.83%	9.86 %	8.58%
Tariff	8.25%	9.86%	11.38%
Annual generation	8.25%	9.86%	11.38%

The results of the sensitivity analysis indicate that none of the input variances put the project at risk of exceeding the benchmark project IRR.

In conclusion, the proposed project activity requires additional revenues from the sale of carbon credits to overcome the risks associated with the investment and to be economically attractive to project developers and surpass or get closer to the benchmark.

Step 3: Barrier analysis

§28 of Tool 01 allows for project proponents to proceed to either Step 2 (Investment analysis) or Step 3 (Barrier analysis). Project proponents may also select to complete both Steps 2 and 3. Accordingly, Step 2 is selected. Step 3 Barrier analysis was not deemed necessary and was not considered in this analysis in accordance with the requirements of the Tool.

Step 4: Common practice analysis

The common practice analysis was carried out as stipulated in Tool 24 version 3.1. This tool outlines a step wise approach, which is described below:

Step 1: Calculate applicable capacity output range as +/- 50% of the total design capacity.

The design capacity for the de Aar Hydra project is 75 MW and therefore the capacity range for this analysis ranges from 37.5 MW to 112.5 MW.

Step 2: Identify similar projects (both VCS and non-VCS) which fulfill all of the following conditions:

Criteria	Applicability to this project	Projects identified
The projects are located in the applicable geographical area;	The applicable area is the geographical borders of South Africa.	All renewable energy projects listed were considered for this analysis.
The projects apply the same measure as the proposed project activity;	The measure applied is renewable energy fed into the South African electricity grid.	5729,7 MW of renewable energy is supplied to the South African grid (extracted from Eskom's website) [9].
The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;	The fuel source is solar with integrated battery storage.	Currently, there is only one other solar with integrated battery storage project supplying to the South African grid, that has already been commissioned, and this project falls outside the capacity range, as it has a capacity of 540 MW.
The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;	Renewable electricity	Solar PV integrated with BESS to provide dispatchable power to the grid
The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;	37.5 MW to 112.5 MW	The capacity of the other project is 150 MW and therefore it falls outside the scope N = 0
The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.		Not applicable as the other project falls outside of the capacity scope.

It can therefore be concluded that $N_{ALL} = 0$.

Steps 4 and 5 of this section are not applicable since $N_{ALL} = 0$

It can therefore be concluded that the proposed project activity is not common practice within South Africa.

3.6 Methodology Deviations

This section is not applicable, as there is no deviation of methodology applied by the Project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as set out in AM tool 07, and the calculation steps are described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y	=	Baseline emissions in year y (tCO ₂ /yr)
$EG_{PJ,y}$ (MWh/yr)	=	Net electricity fed into the grid supplied by the project activity in year y
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO ₂ /MWh)

The results obtained are: 431, 895 t CO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	431, 895	t CO ₂ / yr
$EG_{PJ,y}$	452, 000	MWh/yr
$EF_{grid,CM,y}$	0,96	t CO ₂ /MWh

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$	=	Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EG_{facility,y}$	=	Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 452 000 MWh / yr (from technical specifications).

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with Tool07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

4.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd. This is shown schematically in Figure 7 below.

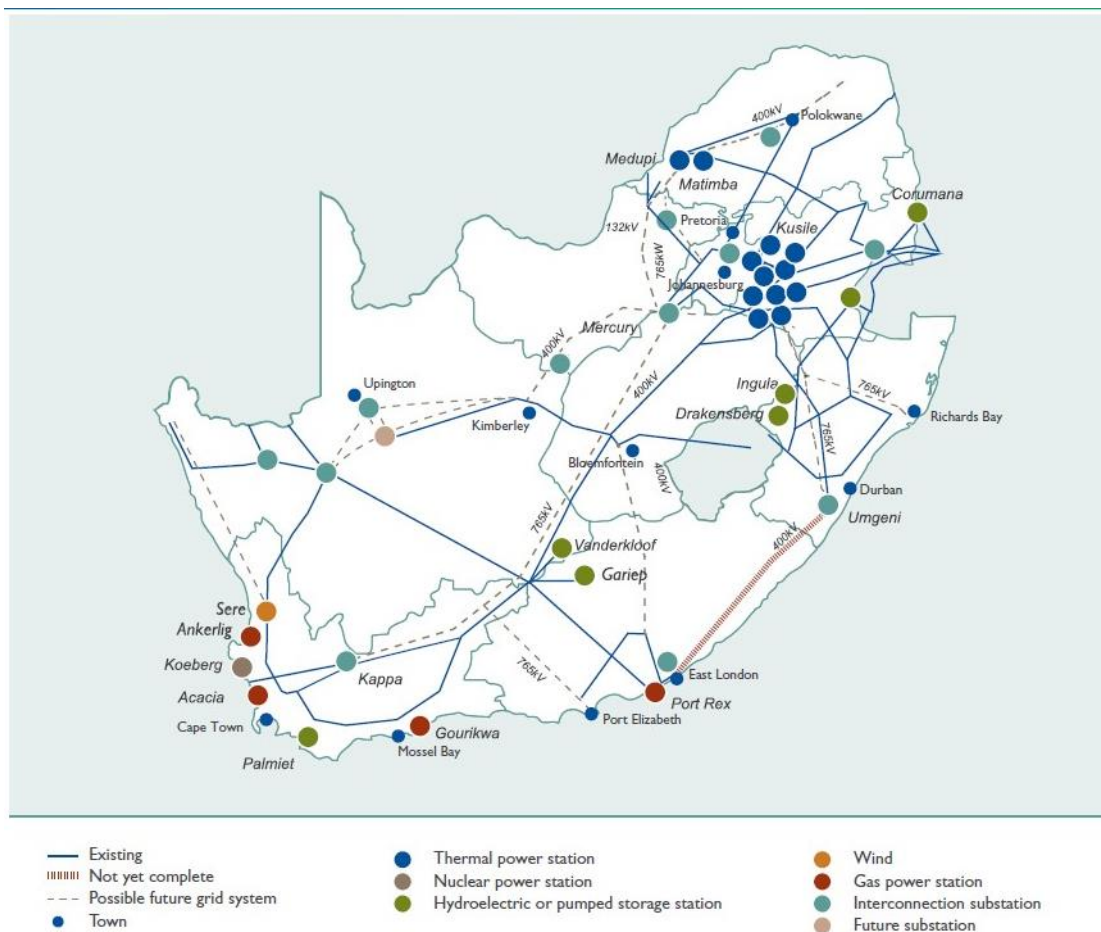


Figure 7: Project electricity system Source: [1]

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country, therefore the grid emission factor is not zero.

4.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in Table 7). were included in this calculation.

4.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (paragraph 38.a in TOOL07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL07 40(a) Approach 1, by Equation 3 below:

$$Share_{LCMR} = average \left[\frac{EG_{LCMR}(y-4)}{total_{y-4}}, \dots, \frac{EG_{LCMR}(y)}{total_y} \right]$$

Equation 3 (Tool07, 40 Equation 1)

Where:

- Share_{LCMR} = Share of the low cost/must run resources (%)
- EG_{LCMRy} = Electricity generation supplied by LCMR plants in year y (MWh)
- total_y = Total electricity supplied to the project electricity system in year y (MWh)
- Y = The most recent year for which data is available

The average share of LCMR plants (share_{LCMR}) was found to be equal to 19,9% over the last 5 years.

The ex-ante option (TOOL07 paragraph 42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2023. Therefore, the generation weighted average of 2021, 2022, and 2023 was used.

The results obtained are shown in Table 5 below

Table 5: LCMR Analysis

Parameter name	Value	Unit
Share _{LCMR} : 2023	23,1%	Per cent
Share _{LCMR} : 2022	21,3%	Per cent
Share _{LCMR} : 2021	19,4%	Per cent
Share _{LCMR} : Average (5 years)	19,9 %	Per cent
EG _{LCMRy} : 2023	49 691	GWh / yr
EG _{LCMRy} : 2022	48 092	GWh / yr
EG _{LCMRy} : 2021	42 495	GWh / yr
total _y : 2023	215 318	GWh / yr
total _y : 2022	226 226	GWh / yr
total _y : 2021	219 423	GWh / yr

4.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including

LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 6 LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
• Gariep • Vanderkloof • Mbashe • First Falls • Ncora • Second Falls	Hydroelectric power stations	Renewable (low generation costs)
• Drakensberg • Ingula • Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
• Acacia • Ankerlig • Gourikwa • Port Rex	Gas and liquid turbine peaking stations	Operated only during hours of highest daily, weekly or seasonal loads (must run)
• Sere	Wind energy stations	Renewable (low generation costs)
• Koeberg	Nuclear power stations	Nuclear listed as LCMR
• Biomass • Concentrated solar power • Gas/liquid fuel • Hydroelectric • Landfill • Solar PV • Wind	Imports and REIPPP plants	Imports or renewable

The power plants used in the OM calculation are shown in Table 7 below. The simple OM was calculated based on the net electricity generation and the CO₂ emission factor of each power unit (TOOL07 paragraph 47, option A), as per Equation 4.

$$EF_{grid,OMsimple,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 4 (Tool 07 48, Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y} = 1.008 \text{ tCO}_2/\text{MWh}$

All values used are shown in Table 7.

Table 7: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
Arnot	10 148 766	9 085 064	7 530 158	1,17	1,16	1,14
Duvha	11 230 400	9 822 489	9 555 266	1,11	1,13	1,13
Hendrina	4 873 506	4 738 179	2 136 620	1,35	1,24	1,20
Kendal	15 106 544	15 139 746	15 750 630	1,02	1,05	1,04
Kriel	11 328 488	12 597 519	10 727 877	1,32	1,28	1,22
Lethabo	21 832 979	22 890 481	21 367 191	1,00	0,97	0,95
Matimba	23 533 126	23 180 336	23 257 630	1,02	1,04	0,99
Majuba	21 857 835	19 823 301	15 540 628	1,19	1,13	1,09
Matla	18 560 297	18 412 664	15 170 449	1,30	1,27	0,99
Tutuka	10 554 111	8 524 715	7 564 337	1,18	1,23	1,22
Camden	2 518 438	6 670 341	7 378 745	1,10	1,16	1,25
Grootvlei	2 992 859	2 445 162	2 416 293	1,21	1,27	1,24
Komati	708 862	608 728	369 330	1,01	1,03	1,06
Kusile	8 553 383	7 992 189	9 119 476	1,01	1,30	1,09
Medupi	19 753 895	22 637 675	23 246 212	0,88	0,92	0,91
Total	183 553 489	184 568 589	171 130 842			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in Equation 5 below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (Tool 07 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO_2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 7 above.

The values used in Equation 5 are shown in Table 8 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports. [1], [10].

Table 8: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO ₂ /GJ)	Emission factor fuel oil (tCO ₂ /GJ)
Arnot	5 663 126	5 186 766	4 293 316	45 179,7	37 792,2	36 433,7	21,43	40,40	0,09	0,08
Duvha	6 581 000	5 732 195	5 428 210	71 179,4	39 550,9	34 435,1	20,39	40,40	0,09	0,08
Hendrina	3 024 527	3 024 981	1 486 626	7 722,1	11 012,5	7 013,8	20,51	40,40	0,09	0,08
Kendal	9 272 316	9 362 475	9 395 998	106 402,9	61 174,9	32 683,4	17,87	40,40	0,09	0,08
Kriel	6 694 760	7 838 445	6 865 127	53 949,1	39 516,2	47 877,9	21,64	40,40	0,09	0,08
Lethabo	15 158 029	16 277 939	15 646 686	14 075,3	12 792,0	9 740,7	14,53	40,40	0,09	0,08
Matimba	12 938 911	13 430 751	13 179 578	14 066,5	9 944,3	13 465,6	19,15	40,40	0,09	0,08
Majuba	12 773 092	11 924 693	9 831 151	140 169,2	149 002,7	89 746,1	19,58	40,40	0,09	0,08
Matla	10 694 976	10 842 908	9 536 046	15 923,0	14 520,0	9 179,0	18,19	40,40	0,09	0,08
Tutuka	6 697 281	5 510 214	4 697 013	64 592,8	68 779,3	124 521,3	19,83	40,40	0,09	0,08
Camden	1 569 620	3 932 224	4 129 344	49 016,8	57 014,4	41 321,3	20,45	40,40	0,09	0,08
Grootvlei	1 938 931	1 544 269	1 518 602	21 199,1	61 318,1	19 380,3	20,01	40,40	0,09	0,08
Komati	402 289	335 833	197 954	1 773,0	1 332,0	2 214,0	19,72	40,40	0,09	0,08
Kusile	5 047 912	4 783 403	5 035 930	10 351,0	534 043,2	32 165,0	19,42	40,40	0,09	0,08
Medupi	9 763 379	11 351 788	11 181 866	23 246,2	22 637,7	19 753,9	19,42	40,40	0,09	0,08

4.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL07 paragraph 72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2023 was 184 165,63 GWh.

The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) was identified as shown in Table 9. The annual generation of $SET_{5 \text{ units}}$ was 35 365,31 GWh in 2023, and this set comprises 19,2% of AEG_{total} .

Table 9: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2023
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Grootvlei	2008/04/30	GWh/yr	2416,29
4	Komati	2009/03/31	GWh/yr	369,33
5	Sere Wind	2015/03/01	GWh/yr	214
TOTAL				35 365,31

The set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} was determined next, and $SET_{\geq 20\%}$ generated 38 379,02 MWh as shown in Table 10 below, comprised 20,8% of AEG_{total} .

Table 10: Set of power plants that contribute 20% to the Eskom fleet power generation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Ankerlig	2007/03/01	GWh/yr	1332,49
4	Gourikwa	2007/07/01	GWh/yr	1681,22
5	Grootvlei	2008/04/30	GWh/yr	2416,29
6	Komati	2009/03/31	GWh/yr	369,33
7	Sere Wind	2015/03/01	GWh/yr	214,00
TOTAL				38 379,02

SET>20% comprised the larger annual electricity generation and therefore forms the sample set SETsample, but it did contain plants that were commissioned more than 10 years ago and therefore step (d) is followed. units that were commissioned more than 10 years ago were excluded from the set (those shaded grey in Table 10), and CDM, Gold Standard (GS), and VCS projects were included until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system [11]. This resulted in a set that comprised 20% of the project electricity system and therefore steps (e) and (f) were not required. The resulting SETsample-CDM group of power units, m, used to calculate the build margin is shown.

Table 11: Plant set for BM Calculation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
Eskom Fleet Plants				
1	Kusile	2017/08/30	GWh/yr	9119,48
2	Medupi	2015/08/22	GWh/yr	23246,21
3	Sere Wind	2015/03/01	GWh/yr	214,00
CDM, VCS and Gold Standard Plants				
4	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	GWh/yr	375,17
5	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	GWh/yr	94,53
6	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	GWh/yr	225,86
7	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	GWh/yr	130,30
8	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	GWh/yr	80,30
9	Longyuan Mulilo 1	2012/01/01	GWh/yr	308,14
10	Longyuan Mulilo 2 de Aar, 144 MW	2012/01/01	GWh/yr	497,38
11	Bokpoort CSP (Concentrating Solar Power) Project South Africa	2016/01/01	GWh/yr	200,00
12	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	GWh/yr	878,44
13	Caledon Wind Farm in South Africa	2012/06/01	GWh/yr	406,01
14	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	GWh/yr	140,45
15	Indwe Wind Project	2012/06/01	GWh/yr	148,09
16	Grid Connected Wind Power Plant in Witberg	2012/02/01	GWh/yr	507,20
17	Red Cap Kouga Wind Farm	2011/11/02	GWh/yr	290,00
TOTAL				4281,85

As the annual electricity generation of this set comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} > 0.2 \times AEG_{total}$), this sample group ($SET_{sample-CDM}$) is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO_2/MWh) for power units m during 2023, which is the most recent year y for which electricity generation data is available, using Equation 6:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 6 (Tool 07 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y} = 0.81$

$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in Table 7.

4.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per Equation 7.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 7 (Tool 07 85 Equation 16)

As per TOOLI07, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM} :

- $w_{OM} = 0.75$
- $w_{BM} = 0.25$

The results obtained are: $EF_{grid,CM,y} = 0.959$

4.2 Project Emissions

As per the approved consolidated Methodology ACM0002, “Grid-connected electricity generation from renewable sources” (Version 22.0) para 40: “For most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where,

PE_y = Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

$PE_{BESS,y}$ = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (tCO₂e/yr)

$PE_{PSP,y}$ = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO₂e/yr)

As the project activity is the installation of a new grid-connected solar Power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore $PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$, $PE_{BESS,y}$, $PE_{PSP,y}$ are equal to zero and thus, $PE_y = 0$.

4.3 Leakage Emissions

As per the applied methodology, no other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g.: extraction, processing, transport etc.) are neglected.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the paragraph 72 of the methodology ACM0002 “Grid-connected electricity generation from renewable sources” Version 22.0 emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \text{ (Equation (17))}$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
30-Sep-2025 to 31-Dec-2025	150,000	0	0	150,000	0	150,000
01-Jan-2026 to 31-Dec-2026	450,000	0	0	450,000	0	450,000
01-Jan-2027 to 31-Dec-2027	450,000	0	0	450,000	0	450,000
01-Jan-2028 to 31-Dec-2028	450,000	0	0	450,000	0	450,000
01-Jan-2029 to 31-Dec-2029	450,000	0	0	450,000	0	450,000
01-Jan-2030 to 31-Dec-2030	450,000	0	0	450,000	0	450,000
01-Jan-2031 to 31-Dec-2031	450,000	0	0	450,000	0	450,000
01-Jan-2032 to 29-Sep-2032	300,000	0	0	300,000	0	300,000
Total	3,150,000	0	0	3,150,000	0	3,150,000

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL07 “Tool to calculate the emission factor for an electricity system”
Value applied	0,959
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. This was done as per equation 16 of TOOL07 “Tool to calculate the emission factor for an electricity system”. Where WOM is equal to 0.75 and WBM is equal to 0.25, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operating margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL07 “Tool to calculate the emission factor for an electricity system” Data Sources: Atmospheric Emission License (EAL) reports – Eskom Data Portal

Value applied	1.008
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom's coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL07 "Tool to calculate the emission factor for an electricity system", based on the net electricity generation and a CO₂ emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	EF _{grid,BM,y}
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 6, as per TOOL07 "Tool to calculate the emission factor for an electricity system" Data source:
Value applied	0.811
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL07 "Tool to calculate the emission factor for an electricity system". An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	Share _{LCMR}								
Data unit	Percent (%)								
Description	Share of the low cost/must run resources								
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [1] and records published on Eskom’s website [10] for each power station.								
Value applied	Annual LCMR (%) <table><tr><td>2021</td><td>2022</td><td>2023</td></tr><tr><td>19,4%</td><td>21,3%</td><td>23,1%</td></tr></table>			2021	2022	2023	19,4%	21,3%	23,1%
2021	2022	2023							
19,4%	21,3%	23,1%							
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 5								
Purpose of data	To calculate baseline emissions, using Equation 3.								
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.								

Data / Parameter	EG _{LCMR,y}							
Data unit	MWh							
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year y							
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [1] and records published on Eskom’s website [10] for each power station.							
Value applied	<table><tr><td>EG_{LCMRy}: 2023</td><td>49 691 000</td></tr><tr><td>EG_{LCMRy}: 2022</td><td>48 092 000</td></tr><tr><td>EG_{LCMRy}: 2021</td><td>42 495 000</td></tr></table>		EG _{LCMRy} : 2023	49 691 000	EG _{LCMRy} : 2022	48 092 000	EG _{LCMRy} : 2021	42 495 000
EG _{LCMRy} : 2023	49 691 000							
EG _{LCMRy} : 2022	48 092 000							
EG _{LCMRy} : 2021	42 495 000							
Justification of choice of data or description of	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from							

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 5
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	total _y		
Data unit	MWh		
Description	Total electricity generation supplied to the project electricity system in year y		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and monthly records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and records published on Eskom's website [10] for each power station		
Value applied	total _y : 2023	215 318	GWh / yr
	total _y : 2022	226 226	GWh / yr
	total _y : 2021	219 423	GWh / yr
	total _y : 2023	215 318 000	
	total _y : 2022	226 226 000	
	total _y : 2021	219 423 000	
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 5		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.		

Data / Parameter	Y
Data unit	Year
Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and records published on Eskom's website [10] for each power station
Value applied	2023
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 5
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in Tool 07.

Data / Parameter	$EF_{grid,OMsimple,y}$
Data unit	tCO ₂ e/MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and records published on Eskom's website [10] for each power station
Value applied	1,.008
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 7
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The simple OM emission factor is calculated as the generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants. using Equation 3 under Option A from section 48 in Tool 07.

Data / Parameter	EG _{m,y}			
Data unit	MWh			
Description	Net electricity delivered to the grid by power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [11] and monthly records published on Eskom’s website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [1] and records published on Eskom’s website [10] for each power station			
Value applied	As per Table 7			
	Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023
	Arnot	10 148 766	9 085 064	7 530 158
	Duvha	11 230 400	9 822 489	9 555 266
	Hendrina	4 873 506	4 738 179	2 136 620
	Kendal	15 106 544	15 139 746	15 750 630
	Kriel	11 328 488	12 597 519	10 727 877
	Lethabo	21 832 979	22 890 481	21 367 191
	Matimba	23 533 126	23 180 336	23 257 630
	Majuba	21 857 835	19 823 301	15 540 628
	Matla	18 560 297	18 412 664	15 170 449
	Tutuka	10 554 111	8 524 715	7 564 337
	Camden	2 518 438	6 670 341	7 378 745

	Grootvlei	2 992 859	2 445 162	2 416 293
	Komati	708 862	608 728	369 330
	Kusile	8 553 383	7 992 189	9 119 476
	Medupi	19 753 895	22 637 675	23 246 212
	Total	183 553 489	184 568 589	171 130 842
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 7			
Purpose of data	To calculate baseline emissions, using Equation 4.			
Comments	The generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in Tool 07.			

Data / Parameter	$EF_{EL,m,y}$
Data unit	tCO ₂ /MWh
Description	CO ₂ emission factor of power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	As per Table 7

	Plant Name	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
	Arnot	1,17	1,16	1,14
	Duvha	1,11	1,13	1,13
	Hendrina	1,35	1,24	1,20
	Kendal	1,02	1,05	1,04
	Kriel	1,32	1,28	1,22
	Lethabo	1,00	0,97	0,95
	Matimba	1,02	1,04	0,99
	Majuba	1,19	1,13	1,09
	Matla	1,30	1,27	0,99
	Tutuka	1,18	1,23	1,22
	Camden	1,10	1,16	1,25
	Grootvlei	1,21	1,27	1,24
	Komati	1,01	1,03	1,06
	Kusile	1,01	1,30	1,09
	Medupi	0,88	0,92	0,91
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 7.			
Purpose of data	To calculate baseline emissions, using Equation 5.			
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.			

Data / Parameter	$FC_{i,m,y}$
Data unit	Tonnes
Description	Amount of fuel type i consumed by power unit m in year y

Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [12] for each power station to fulfil the Atmospheric emission license (EAL) requirements

Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL) requirements

Value applied

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021
Arnot	5 663 126	5 186 766	4 293 316	45 179,7	37 792,2	36 433,7
Duvha	6 581 000	5 732 195	5 428 210	71 179,4	39 550,9	34 435,1
Hendrina	3 024 527	3 024 981	1 486 626	7 722,1	11 012,5	7 013,8
Kendal	9 272 316	9 362 475	9 395 998	106 402,9	61 174,9	32 683,4
Kriel	6 694 760	7 838 445	6 865 127	53 949,1	39 516,2	47 877,9
Lethabo	15 158 029	16 277 939	15 646 686	14 075,3	12 792,0	9 740,7
Matimba	12 938 911	13 430 751	13 179 578	14 066,5	9 944,3	13 465,6
Majuba	12 773 092	11 924 693	9 831 151	140 169,2	149 002,7	89 746,1
Matla	10 694 976	10 842 908	9 536 046	15 923,0	14 520,0	9 179,0
Tutuka	6 697 281	5 510 214	4 697 013	64 592,8	68 779,3	124 521,3
Camden	1 569 620	3 932 224	4 129 344	49 016,8	57 014,4	41 321,3
Grootvlei	1 938 931	1 544 269	1 518 602	21 199,1	61 318,1	19 380,3
Komati	402 289	335 833	197 954	1 773,0	1 332,0	2 214,0
Kusile	5 047 912	4 783 403	5 035 930	10 351,0	534 043,2	32 165,0
Medupi	9 763 379	11 351 788	11 181 866	23 246,2	22 637,7	19 753,9

As per Table 8

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.

Data / Parameter	NCV _{i,y}																																																		
Data unit	GJ/tonne																																																		
Description	Net calorific value of fuel type i in year y																																																		
Source of data	DFFE for coal [13] IPCC default factors for heavy fuel oil , [14]																																																		
Value applied	For coal as per plant in Table 8																																																		
	<table><tr><th>Plant Name</th><th>Net calorific value coal (GJ/t)</th><th>Net calorific value fuel oil (GJ/t)</th></tr><tr><td>Arnot</td><td>21,43</td><td>40,40</td></tr><tr><td>Duvha</td><td>20,39</td><td>40,40</td></tr><tr><td>Hendrina</td><td>20,51</td><td>40,40</td></tr><tr><td>Kendal</td><td>17,87</td><td>40,40</td></tr><tr><td>Kriel</td><td>21,64</td><td>40,40</td></tr><tr><td>Lethabo</td><td>14,53</td><td>40,40</td></tr><tr><td>Matimba</td><td>19,15</td><td>40,40</td></tr><tr><td>Majuba</td><td>19,58</td><td>40,40</td></tr><tr><td>Matla</td><td>18,19</td><td>40,40</td></tr><tr><td>Tutuka</td><td>19,83</td><td>40,40</td></tr><tr><td>Camden</td><td>20,45</td><td>40,40</td></tr><tr><td>Grootvlei</td><td>20,01</td><td>40,40</td></tr><tr><td>Komati</td><td>19,72</td><td>40,40</td></tr><tr><td>Kusile</td><td>19,42</td><td>40,40</td></tr><tr><td>Medupi</td><td>19,42</td><td>40,40</td></tr></table>	Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Arnot	21,43	40,40	Duvha	20,39	40,40	Hendrina	20,51	40,40	Kendal	17,87	40,40	Kriel	21,64	40,40	Lethabo	14,53	40,40	Matimba	19,15	40,40	Majuba	19,58	40,40	Matla	18,19	40,40	Tutuka	19,83	40,40	Camden	20,45	40,40	Grootvlei	20,01	40,40	Komati	19,72	40,40	Kusile	19,42	40,40	Medupi	19,42	40,40		
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Kusile	19,42	40,40																																																	
Medupi	19,42	40,40																																																	

Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in Tool 07 is applied.

Data / Parameter	AEG_{total}
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	184 165,63
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 7.
Purpose of data	To calculate baseline emissions
Comments	No additional comments

Data / Parameter	$SET_{5\ units}$
Data unit	Dimensionless
Description	The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil

	the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01) Komati (2009/03/31) Grootvlei (2008/04/40)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 9
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20,8% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	35 365,31
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 9
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 19,2% of AEG_{total} in 2023.

Data / Parameter	$SET_{\geq 20\%}$
Data unit	Dimensionless
Description	The set of power units, excluding power units registered as CDM project activities, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Ankerlig (2007/03/01) Gourikwa (2007/07/01) Grootvlei (2008/04/30) Komati (2009/03/31) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20,8% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set \geq 20\%}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual

	records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	38 379,02
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20,8% of AEG _{total} in 2023.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, and exclude power units that started to supply electricity to the grid more than 10 years ago
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in . Table 11 and Table 10
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 32,579,69 GWh in 2023, comprising 17,7% of AEG _{total} in 2023.

Data / Parameter	$AEG_{set\ sample}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	32,579,69
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in. Table 11 and Table 10
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of AEG_{total} in 2023.

Data / Parameter	$SET_{sample-CDM}$
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\ units}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, excludes power units that started producing electricity more than 10 years ago, and including power units registered as CDM project activities, that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)

Value applied	Shown in . Table 11
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in . Table 11
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 36 861,54 GWh in 2023, comprising 20,0% of AE_{total}

Data / Parameter	$AE_{SET-sample-CDM}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AE_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [1] and monthly and annual records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	36 861,54
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in . Table 11
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated $\geq 20,0\%$ of AE_{total} in 2023.
Data / Parameter	w_{OM}

Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor
Source of data	Tool07 paragraph 86 (a)
Value applied	75%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM Tool 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 75% for the first crediting period and for subsequent crediting periods.

Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	Tool07 paragraph 86 (a)
Value applied	25%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM Tool 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 25% for the first crediting period and for subsequent crediting periods.

Data / Parameter	BE_y
Data unit	tCO _{2e} /yr
Description	Emission reductions in year y

Source of data	Quantified using Equation 1
Value applied	To be determined ex-post.
Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.
Purpose of data	To calculate baseline emissions, using Equation 1
Comments	No additional comments

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$														
Data unit	MWh/Year														
Description	Quantity of net electricity generation supplied by the project to the grid in year y														
Source of data	Electricity reading meters														
Description of measurement methods and procedures to be applied	Electricity generated is metered automatically, recorded electronically on the EMS and SCADA systems, recorded monthly, following the Plant's Standard Operating Procedures (SOPs), and checked against the Eskom meter.														
Frequency of monitoring/recording	Continuous monitoring and monthly reporting														
Value applied	Ex-ante values for the first crediting period: <table border="1"> <tr> <td>30 September-2025 to 31-December-2025</td><td>150,000</td></tr> <tr> <td>01-January-2026 to 31-December-2026</td><td>450,000</td></tr> <tr> <td>01-January-2027 to 31-December-2027</td><td>450,000</td></tr> <tr> <td>01-January-2028 to 31-December-2028</td><td>450,000</td></tr> <tr> <td>01-January-2029 to 31-December-2029</td><td>450,000</td></tr> <tr> <td>01-January-2030 to 31-December-2030</td><td>450,000</td></tr> <tr> <td>01-January-2031 to 31-December-2031</td><td>450,000</td></tr> </table>	30 September-2025 to 31-December-2025	150,000	01-January-2026 to 31-December-2026	450,000	01-January-2027 to 31-December-2027	450,000	01-January-2028 to 31-December-2028	450,000	01-January-2029 to 31-December-2029	450,000	01-January-2030 to 31-December-2030	450,000	01-January-2031 to 31-December-2031	450,000
30 September-2025 to 31-December-2025	150,000														
01-January-2026 to 31-December-2026	450,000														
01-January-2027 to 31-December-2027	450,000														
01-January-2028 to 31-December-2028	450,000														
01-January-2029 to 31-December-2029	450,000														
01-January-2030 to 31-December-2030	450,000														
01-January-2031 to 31-December-2031	450,000														

	01-January-2032 to 29-September-2032	300,000
	Total	450,000
Monitoring equipment	Electricity meter, (to be confirmed during installation and commissioning) In-access SCADA and EMS systems which will operate and run the grid electrical feed.	
QA/QC procedures to be applied	Internal Operational QA/QC for meter reading via the SCADA system have been developed and will meet globally acceptable industry norms. Specifics of the QA/QC will be codified during the implementation and commission of the plant and included at verification.	
Purpose of data	Calculation of project emissions	
Calculation method	Net electricity supplied to the grid by the project plant is directly measured and reported, not calculated.	
Comments	No additional comments	

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Project emissions in year y
Source of data	Quantified using Equation 8
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	No additional QA/QC would be applied to this equation

Purpose of data	Calculation of project emissions
Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoirs of hydro power plants; therefore, project emissions for this renewable energy power generation will be zero.

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	To be calculated ex-post. Ex-ante value is 450 000 per annum
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	No additional QA/QC would be applied to this equation
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emissions reductions from the proposed



project activity during the crediting period, using available ex-ante values. These will be revised during each verification using the actual data recorded (where applicable).

5.3 Monitoring Plan

Understanding that the project is still in design and implementation phase and has not been constructed and final equipment and instrumentation is still to be installed and implemented. This is the best understanding at the time of validation and will be amended and corrected in an “as-built” version of the plant.

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

The project will have a PV forecasting system that will provide PV power forecasts to the Energy Management System (EMS) via the Facility Supervisory Control and Data Acquisition (SCADA). The Facility SCADA will share plant data and forecast data with a 3rd Party system in order to receive a generation forecast, declaration schedule, and availability notices from the 3rd Party system. The Facility SCADA will relay the generation forecast to the EMS and Network Service Provider (NSP).

The below diagram indicates the data exchange with the 3rd Party Generation Forecast provider(s).

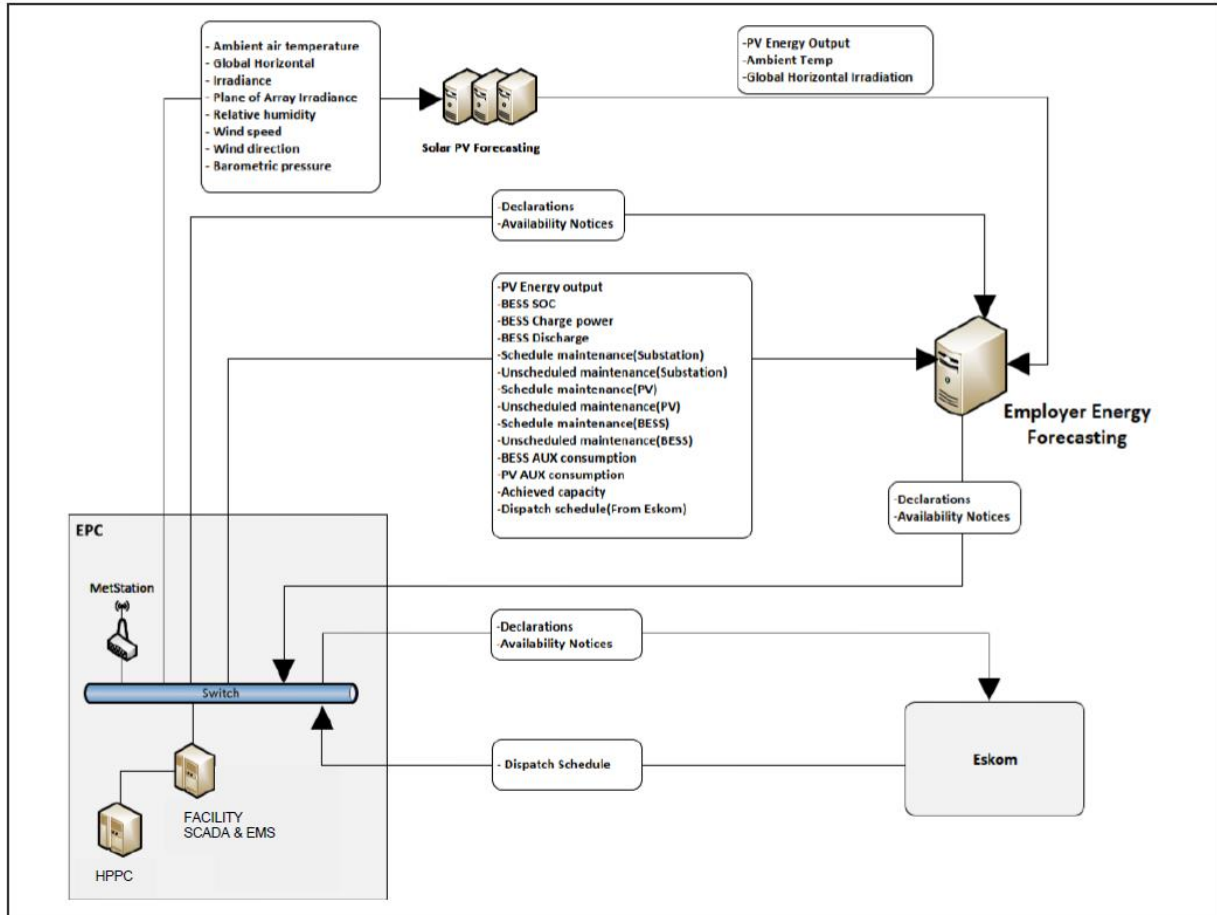


Figure 8: Data exchange schematic for Solar PV forecasting

The EMS will calculate and produce a solar PV power generation forecast for 72-hours based on historical generation data as well as plant weather data. The EMS will determine a 6-day power generation forecast from the plant statuses, availability and the PV plant forecasting system. The EMS will also accept the generation forecast from the Employer's system and produce a report to indicate the variances between the two forecasts.

6-day forecasting (external)

A third-party forecasting system will be used to forecast the following parameters over a 6-day period:

- Ambient temperature (10-min resolution)
- Wind speed (10-min resolution)
- Global horizontal irradiance (10-min resolution)
- PV power production (10-min resolution)

The forecast will be automatically loaded to the EMS daily in CSV or XML format.

72-hour Forecasting (internal)

The EMS will use the imported forecast data in conjunction with the measured values in the facility to produce a 72-hour forecast of the following parameters:

- Ambient temperature (1-hour resolution)
- Wind speed (1-hour resolution)
- PV power production (15-min resolution)

The EMS will use the wind speed forecast to determine if there are any high wind speeds predicted that could be considered dangerous to modules and that could potentially lead to the modules requiring to stow in order to mitigate the risk of damage. The EMS will raise an alarm and adjust the PV power production forecast accordingly.

The EMS will cross-reference the PV power production forecast with the available smart decision-making system to determine the anticipated amount of power available to charge the BESS. The EMS will further determine if the PV power forecast, in conjunction with the BESS current SoC and forecast surplus power will be sufficient to meet the available dispatch instruction in its entirety.

The EMS will also provide updated generation forecasts one (1) and six (6) hours before the availability period, this generation forecast will be used to evaluate if revised availability notices must be sent to Eskom before the start of the availability period.

If the EMS determines that the dispatch instruction will not be met, it will raise an alarm.

As per the requirements of the PPA, the project shall include an accurate and up-to-date log of Operations, updated daily. The Project Data should include, but not be limited to, the following information, with records of:

- the Energy Output and the Reactive Energy Output.
- changes in Operating status during the day.
- the number of Outages or Deratings in the day.
- any unusual conditions found during Maintenance inspections.
- information relating to the performance of the Project and data obtained from the meters.

Measurement data will be managed by the project's DCS. The measuring equipment will be installed in three steps:

- Calibration under standard temperature and pressure conditions
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data will be stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment will strictly comply with their specified maintenance plans and calibration certificates will be kept on record.

As per the applied methodology, electricity generation will be measured onsite at the outgoing substation, adjacent to the generating turbines. This will be monitored on a continuous basis, with a monthly report being aggregated and reported upon. A check meter, owned by the national utility (Eskom) located at the Eskom Switching Stations which serves as the grid tie-in point. Meters at both locations will have a back-up meter installed as a failsafe.

The meters on site, will be calibrated in accordance with the manufacturer's specifications, and will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records will be kept on site. Calibrations will be conducted by competent technicians under SANAS accreditation. The applicable national standard to be applied is SANS 474. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years and less than 10MVA as being every 10 years.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a dedicated data and instrumentation team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibrations and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS approved organisations will conduct calibrations.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the EMS system. This will be checked against the Eskom meter at the independent switching stations. Any discrepancies will immediately trigger fault finding and corrective action procedures. All reported electricity generation values will be checked by both project personnel and Eskom personnel prior to billing and payment.



APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No Commercially sensitive information was included in this version of the PD

APPENDIX 2: BIBLIOGRAPHY

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